

Rates and Unit Rates

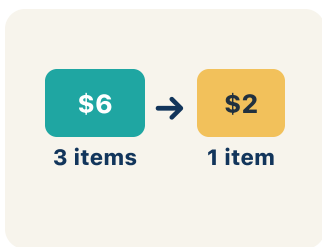
Lesson 4-1

Name: _____ Date: _____ Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.



\$12 for 4 games is a rate

Rate

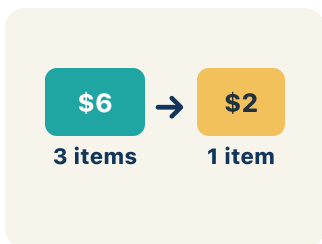
A ratio comparing two amounts with different units, like miles per hour.



\$3 per 1 game

Unit Rate

A rate for just 1 of something, like cost for 1 item.



60 miles per hour means 60 miles for every 1 hour

Per

For each one. Example: 5 dollars per book.

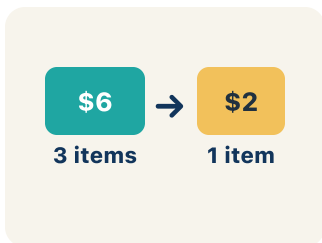


3 : 2

5 to 3 or 5:3

Ratio

A way to compare two amounts.



Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil — Pack B is the better buy

Better Buy

The choice that costs less for each item.



$6 \div 3 = 2$

\$12 ÷ 4 games = \$3 per game

Divide

Splitting a total into equal groups.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find a unit rate to compare prices and decide the better buy.

1 Notice

You're comparing ticket prices at different game booths in the arcade.

2 Key idea

I can find a unit rate to compare prices and decide the better buy.

3 Words I'll use

Rate

Unit Rate

Per

Ratio

Better Buy

Divide



Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A booth charges \$4.50 for 9 games. What is the unit rate?

- A. \$0.50 per game
- B. \$0.45 per game
- C. \$2.00 per game
- D. \$4.50 per game

Step 1

$$\$4.50 \div 9 = \$0.50 \text{ per game.}$$



Answer: A. \$0.50 per game

 We do — together

Solve this one with your class using the same steps.

Problem: A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

- A. 30 miles per gallon
- B. 6 miles per gallon
- C. 180 miles per gallon
- D. 36 miles per gallon

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A pack of 8 markers costs \$6.00. What is the cost per marker?

- A. \$0.75
- B. \$0.80
- C. \$1.33
- D. \$0.60

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Booth A charges \$3 for 5 games and Booth B charges \$5 for 8 games. Just by looking, which booth seems like the better deal, and what makes it hard to be sure?

Level 1 support

Start here: It is hard to compare because the booths have different numbers of games.

 **Need a hint?**

Sentence starters (optional)

Booth ____ looks better because ____.

El puesto ____ se ve mejor porque ____.

It is hard to compare because the ____ are different.

Es difícil comparar porque los/las ____ son diferentes.

Word bank: rate per ratio compare

Listen for: Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

Level 2

Why can you not just pick the booth with the lower total price of \$3? Explain what a fair comparison actually needs.

- You cannot just use the lower price because ____.
- A fair comparison needs ____ so that ____.

EXPLORE

How did you find the cost per game (the unit rate) for each booth, and what did you divide?

Level 1 support

Start here: I divided the total cost by the number of games to get the cost for one game.



Need a hint?

Sentence starters (optional)

The unit rate is ____ per one ____.

La tasa unitaria es ____ por un ____.

I divided ____ by ____ to get the cost per game.

Dividí ____ entre ____ para obtener el costo por juego.

Word bank: unit rate divide per rate cost

Listen for: Students explain they divided total cost by number of games (e.g., $\$5.00 \div 8 = \0.625) and state the result as a cost per one game.

Level 2

Two booths in your table had different totals but the same unit rate. How is that possible, and what does it tell you about the deals?

- They can have the same unit rate because ____.
- Even though the totals differ, the deals are ____ because ____.

CONNECT

The snack bar sells popcorn as 3 cups for \$2.25 or 8 cups for \$5.60. How do unit rates help you decide the better buy?

Level 1 support

Start here: I find the price per cup for each bag and pick the lower one.



Need a hint?

Sentence starters (optional)

The ____ bag is the better buy because its unit rate is \$ ____ per cup.

La bolsa ____ es la mejor compra porque su tasa unitaria es \$ ____ por taza.

I divided ____ by ____ to find the price per cup.

Dividí ____ entre ____ para hallar el precio por taza.

Word bank: (unit rate) (better buy) (divide) (per) (cost)

Listen for: Students compute \$0.75/cup and \$0.70/cup, and choose the large bag because the lower unit rate means each cup costs less.

Level 2

A friend says the large bag must be the better buy because it costs less per cup AND gives more popcorn. Is 'more popcorn' a fair reason to call it the better buy? Justify using the unit rate.

- 'More popcorn' is or is not a fair reason because ____.
- The real reason it is the better buy is ____.

Write About the Math

The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

1. Kernel Sentence subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.

Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.

Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rate matters in math

Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rate matters in math because ____.

Tasa importa en matemáticas porque ____.

but
pero

Rate matters in math, but ____.

Tasa importa en matemáticas, pero ____.

so
entonces

Rate matters in math, so ____.

Tasa importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question
Pregunta

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Booth ____ **looks better because** ____.

El puesto ____ *se ve mejor porque* ____.

It is hard to compare because the ____ **are different.**

Es difícil comparar porque los/las ____ *son diferentes.*

The unit rate is ____ **per one** ____.

La tasa unitaria es ____ *por un* ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. Which is the better unit rate for typing speed: 240 words in 4 minutes or 300 words in 6 minutes?

- A. 240 words in 4 minutes
- B. 300 words in 6 minutes
- C. They are equal
- D. Cannot tell

Show your work:

2. A faucet fills 15 liters in 5 minutes. How many liters per minute?

- A. 3
- B. 75
- C. 5
- D. 10

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of \$____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 30 miles per gallon — $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon}$.
2. **You Try (notes):** A. $\$0.75 - \$6.00 \div 8 = \$0.75 \text{ per marker}$.
3. **Try It 1:** A. 240 words in 4 minutes — $60 \text{ words/min vs } 50 \text{ words/min}$; 240 in 4 minutes is faster.
4. **Try It 2:** A. $3 - 15 \div 5 = 3 \text{ liters per minute}$.
5. **Exit Ticket:** A. 6 for \$1.50 — \$0.25 each — $\$1.50 \div 6 = \0.25 per pencil . $\$2.80 \div 10 = \0.28 per pencil . $\$0.25 < \0.28 , so 6 for \$1.50 is the better buy.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).